

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

MI-802 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Mining Machinery - III
Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.
ii) All questions carry equal marks.

1. With a neat diagram describe the construction and working of an electrical coal drilling machine. Also explain various drill bits.
2. Describe the constructional features of a Double Ended Ranging Drum Shearer with a suitable diagram.
3. Describe Rotary screw type of compressor with suitable sketch.
4. Explain :
 - i) Intrinsically safe apparatus
 - ii) Flame proof enclosure
 - iii) Underground cables
 - iv) Circuit breaker.
5. What is Gate End Box? What are the various safety and protection devices provided in the Gate End Box?
6. What is Sumping? Explain any one method with neat sketch. How sumping affects the operational efficiency of shearer?
7. What are the cutter loader machines used on Longwall and Shortwall Mining? Give a detailed illustration of the cutting process of Shearer. Give a neat diagram also.

OR

What is Flame Leak proof Apparatus? Discuss the design features of a Flame Leak proof Apparatus.

8. Describe the constructional features of any two
 - i) LHD
 - ii) Drill Jumbo
 - iii) AFC
